higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE (VOCATIONAL)

## PROCESS TECHNOLOGY
## NQF LEVEL 4

## 27 FEBRUARY 2019

**This marking guideline consists of 5 pages.**

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## QUESTION 1

1.1 1.1.1 The product that remains after evaporation has taken place. (2)

1.1.2 It is a separation process that separates two immiscible liquids by making use of gravity and density differences between the mixtures. (2)

1.2 1.2.1 The liquid scrubs the gas as it moves through the system. The gas is absorbed by the liquid as a result of the contact between the gas and the liquid. (2)

1.2.2 In the process of breaking the gas into tiny bubbles inside a continuous liquid, the Tray columns allow for internal cooling to take place. (2)

1.3 It is highly efficient but cheap to run.
A wide variety of packing materials can be used.
It offers low maintenance and trouble free operation. (3)

1.4 The solvent extraction process is a form of absorption in which solid metals are removed from the liquid solution. √ Organic solvent is always used in order to extract the metals from the liquid solution. √ It involves the mixing of the liquid-metal with a reagent. √ The metal will react with the reagent to form a chemical compound. √ The reagent is selected based to form a chemical compound with the metal that will be more soluble in the organic solvent than in the substance it originally came from. √ Then the metal will be extracted from the liquid solution and transferred to the organic solvent. √ Once the metal is transferred to the organic solution, it is mixed with another liquid. √ This new liquid will contain substances designed to break the chemical compound back into the reagent and metal components. √ The continuous exposure of the liquid-metal solution to this process, will lead to the formation of a product with a much higher concentration of the metal. √ All unwanted molecules and other compounds will not be present in the recovered liquid-metal solution since they will not be transferred to the organic solution. √ (10)

1.5 The agitator has blades which are lying flat in a horizontal orientation. √√ The blades of a thickener are placed in a vertical orientation. √√ (4)

1.6 1.6.1 The formation of substances or build-up of organisms inside the exchange tubes. (2)

1.6.2 Leads to reduced efficiency in decantation and may produce products containing excessive amounts of contaminants. (2)

1.6.3 By cleaning the equipment continuously. (1)
**[30]**

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## QUESTION 2

2.1 It is a fully defined space that is closed off to its surroundings by a boundary that prohibits any exchange of mass or energy. (2)

2.2 2.2.1 $V = (\pi \times r^2 \times h) / 3$ (3)

2.2.2 $V = \pi \times r^2 \times h$ (3)

2.3 Matter (mass) cannot be created or destroyed. It can however change from one form to another. (3)

2.4 A process that involves the separation or combination of various atomic elements. (2)

2.5 One mole always has the same number of particles which is given by the Avogadro's number, √ i.e.1 mole = 6,0221 x $10^{23}$ particles. √√ (3)

2.6 2.6.1 3000 L = 3 kL √ (1)

2.6.2 900 mg = 900 000 g √√ (2)

2.7 2.7.1 $Ca(OH)_2 + K_2CO_3 \longrightarrow CaO$ √ + 2KOH √ + $CO_2$ √ (3)

2.7.2 $Zn + 2HCl \longrightarrow ZnCl_2$ √ + $H_2$ √ (2)

2.8 2.8.1 $M_{Na2CO3} = (23 \times 2) + 12 + (16 \times 3)$
$= 46 + 12 + 48$ √
$= 106$ g/mol √ (2)

2.8.2 $M_{Al2(CO3)3} = (27 \times 2) + (12 \times 3) + (48 \times 3)$
$= 54 + 36 + 144$ √
$= 234$ g/mol √ (2)

2.8.3 $M_{HCl} = 1 + 35$ √
$= 36$ g/mol √ (2)
**[30]**

## QUESTION 3

3.1 Energy possessed by an object due to its motion relative to the ground. √√ (2)

$E_K = ½ \times m \times v^2$ √√ (2)

3.2 $^0C = K - 273.15$ √
$= 940\ K - 273.15$ √
$= 666.85\ ^0C$ √√ (4)

3.3 $^0C = 5/9\ (^0F - 32)$ √
$= 5/9\ (230 – 32)$
$= 5/9\ (198)$ √
$= 0.555 \times 198$
$= 110\ ^0C$ √

Since $1\ K = 1\ ^0C$ √

and $\Delta T = 110\ ^0C$ √

$C = Q / \Delta T$ √
$0.79 = Q / 110\ ^0C$ √
$Q = 0.79 \times 110$
$= 86.9\ KJ$ √ (8)

3.4 Solids: Particles are closely packed together and cannot move around but tend to vibrate while remaining in position. (2)

Liquids: Particles are spaced closely and do not vibrate, but move around. (2)

3.5 3.5.1 Boiling point (2)

3.5.2 Melting point (2)

3.5.3 Condensation point (2)

3.6
- It is a form of heat exchange where energy is carried from one substance to another in the form of electromagnetic waves. (2)
- The sun radiates electromagnetic waves through the vacuum of space in order to heat our planet. (2)

**[30]**

**QUESTION 4**

4.1
- Thermal insulators
- Soft drink industry
- Cable insulators
- Packing industry √

(4 × 1) (4)

4.2
- Cellulose is the wall of plant cells which is made up of carbohydrates.
- Lignin is a special polymer that combines with cellulose to form the hard cell walls of a plant. (2 × 2) (4)

4.3
- Open-die forging
- Closed-die forging (2 × 1) (2)

**[10]**

**TOTAL : [100]**

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